

WEIMAR, Germany

WMO: 105550

Lat: 50.983N

Lon: 11.317E

Elev: 879

StdP: 14.23

Time Zone: 1.00 (EUC)

Period: 97-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	16.8	7.8	8.5	15.6	12.3	10.6	18.8	32.2	45.9	29.3	44.8	3.5	80	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.5	85.9	65.7	82.1	64.0	78.8	63.1	67.5	80.3	66.1	77.9	64.7	74.8	5.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	91.0	69.9	62.2	86.4	68.8	60.9	82.4	68.2	32.4	80.0	31.3	77.6	30.2	74.5	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	21.2	17.9	DB	5.2	92.7	5.0	3.0	1.6	94.9	-1.3	96.6	-4.0	98.3	-7.6	100.5	(4)
			WB	5.6	71.9	5.7	2.4	1.5	73.6	-1.8	75.0	-5.0	76.3	-9.2	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	32.7	35.9	40.8	47.5	56.5	61.0	64.3	64.5	57.8	49.6	40.7	34.7
	DBStd	13.30	9.27	8.64	7.35	6.75	5.99	6.30	5.85	5.57	5.99	7.28	6.65	7.66
	HDD50	2229	537	398	294	124	13	2	0	0	6	96	285	474
	HDD65	6048	1001	814	749	526	272	151	86	78	227	477	729	938
	CDD50	1829	1	4	9	48	213	331	443	450	238	85	6	1
	CDD65	173	0	0	0	0	8	30	64	62	9	0	0	0
	CDH74	2027	0	0	0	5	156	374	662	683	144	3	0	0
	CDH80	568	0	0	0	0	29	87	217	217	18	0	0	0
Wind	WSAvg	6.4	7.6	8.9	7.5	6.4	5.6	5.3	5.6	5.2	5.5	7.0	5.8	7.1
Precipitation	PrecAvg	23.7	1.2	1.2	1.6	1.5	2.5	2.6	3.3	2.6	2.1	1.6	1.9	1.6
	PrecMax	31.3	2.6	2.3	4.0	3.9	6.7	4.2	5.9	6.9	4.8	3.9	5.0	3.6
	PrecMin	15.6	0.2	0.1	0.2	0.3	0.4	0.6	0.8	0.8	0.4	0.6	0.0	0.4
	PrecStd	3.8	0.6	0.6	0.9	0.8	1.3	0.8	1.3	1.3	0.9	0.8	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	57.8	64.1	74.1	83.3	87.0	90.7	91.6	82.6	71.8	59.5	54.9
		MCWB	47.8	48.7	51.5	56.9	64.3	66.7	65.5	65.2	62.4	60.3	52.4	49.4
	2%	DB	50.4	54.3	59.7	68.7	78.5	82.4	85.9	85.5	78.6	66.4	54.8	51.1
		MCWB	45.1	47.2	49.4	54.7	61.6	64.9	65.6	65.1	61.3	57.7	49.8	46.6
	5%	DB	48.1	51.7	55.8	64.4	73.8	78.3	81.4	81.3	74.2	63.0	52.3	47.9
		MCWB	44.0	46.1	47.9	52.7	59.8	64.2	64.5	63.3	60.8	55.5	47.9	44.1
	10%	DB	44.6	48.3	52.1	60.5	69.2	74.5	76.7	77.2	69.7	60.0	49.9	45.2
		MCWB	41.1	43.7	46.3	50.6	57.1	61.8	62.9	62.8	59.4	54.3	46.4	41.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	49.5	53.4	59.2	65.9	69.9	69.2	68.9	65.3	61.5	53.6	50.2
		MCDB	53.5	55.0	61.6	70.1	80.0	82.1	81.7	84.3	76.2	69.3	57.2	54.3
	2%	WB	45.8	48.3	50.6	56.4	62.9	67.2	67.2	67.2	63.1	58.5	50.5	46.8
		MCDB	49.6	53.8	57.2	66.0	73.7	78.7	79.5	80.7	73.3	65.2	54.0	50.8
	5%	WB	44.0	46.4	48.7	54.0	61.2	65.4	65.8	65.5	61.6	56.5	48.5	44.3
		MCDB	48.0	51.0	54.4	61.8	71.5	76.1	77.0	76.3	71.1	61.6	51.7	47.5
	10%	WB	41.1	43.9	46.6	51.6	59.0	63.4	64.3	63.8	60.2	54.9	46.6	42.0
		MCDB	44.4	48.0	51.9	58.8	67.4	72.1	74.0	73.6	68.4	59.2	49.7	44.9
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.8	13.6	18.0	19.5	19.6	18.7	19.5	18.0	13.7	9.6	8.5
		MCDBR	11.2	14.4	20.4	25.8	28.5	26.9	27.3	28.2	26.4	19.1	12.9	10.3
		MCWBR	8.2	9.3	12.1	13.3	13.5	12.2	9.7	10.2	11.5	11.0	8.8	8.1
	5% WB	MCDBR	11.0	11.8	17.4	22.3	24.0	24.9	22.6	22.9	21.2	16.7	11.6	9.9
		MCWBR	8.5	8.0	11.1	12.1	12.1	12.0	9.2	9.5	10.1	10.4	8.6	8.1
Clear-Sky Solar Irradiance	taub	0.304	0.326	0.365	0.403	0.407	0.404	0.419	0.412	0.377	0.351	0.327	0.299	
	taud	2.415	2.376	2.296	2.228	2.246	2.276	2.239	2.281	2.363	2.429	2.436	2.408	
	Ebn at Noon	230	254	263	265	269	271	264	259	256	242	218	212	
	Edh at Noon	19	26	34	40	41	40	41	38	31	24	19	17	
All-Sky Solar Radiation	RadAvg	239	464	805	1282	1544	1677	1583	1361	981	571	267	180	
	RadStd	37	67	103	161	180	141	199	136	133	68	36	22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page